Michael Boiman

Quality Engineering & AI Automation Expert • 4+ Years AI Development • 20+ Years Quality Assurance

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- BKS-Lab Homepage
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- BKS-Lab Organization

-80%
weniger Rückfragen
durch Quality-Monitoring

96%
Qualität nach Migration
vorher 30%

80%
KI-gestützte Tests
auto-generiert

Quality Engineering & AI Competencies

Quality Engineering & AI Automation
Quality-first automation: Email bots, invoice processing, workflow automation with 80% workload reduction, measurable cost savings, processing time from hours to seconds

AI Quality Engineering Excellence
80% auto-generated tests & test cases, IDE integration, Go/No-Go in minutes instead of hours, 50% less development time, 80% fewer inquiries

Next-Gen AI Protocols & Agentic Systems
MCP & A2A protocol implementation, Google Agent Development Kit (ADK), multi-agent orchestration, JSON-RPC, secure agent-to-agent communication, enterprise-ready

RAG Systems & Enterprise NLP
Vector embeddings, LangChain, Q&A bots, spaCy/NLTK, 95% accuracy, EU standards, contextual intelligence

Multi-Cloud AI Architecture
Azure OpenAI, Google Cloud AI, serverless MLOps, auto-scaling, real-time monitoring, production-grade deployment

Transparency & Business Intelligence
Real-time dashboards, quality visualization, end-to-end monitoring, KPI automation, single source of truth

Career Profile

Quality Engineer and AI Automation Expert with 20+ years in software quality assurance and 4 years of specialized AI development. My focus: transforming business processes through measurable automation — from test generation to invoice processing.

In enterprise projects (DB Vertrieb, DVAG, TÜV Süd), I built quality monitoring systems that cut Go/No-Go decisions from hours to minutes, introduced AI-driven test automation that generates 80% of test cases automatically, and validated legacy migrations 24/7 — improving quality from 30% to 96%.

Today I combine quality engineering with AI architecture: MCP and A2A protocols, multi-agent systems, e-invoicing platforms to EU standards, and a 40+ skill automation ecosystem for enterprise workflows.

TECHNICAL SKILLS

Quality Engineering | Playwright | Cucumber/Gauge | Python | LLMs (Claude/Gemini/GPT/Copilot) | MCP & A2A Protocols | Google ADK | LangChain | Peppol/E-Invoicing | CI/CD | Kubernetes | Azure Functions | Grafana | Elasticsearch | REST APIs | Docker | JMeter | Gatling

EDUCATION

Diploma in Computer Science (UAS)
Cologne University of Applied Sciences, Gummersbach
2000 – 2005

ISTQB Certified Tester
2005

Certified Dynatrace Diagnostics Basic Training
2025

LANGUAGES

German	Native
English	Professional
Russian	Conversational
Hebrew	Conversational

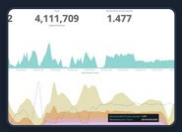
ADDITIONAL QUALIFICATIONS

ISTQB Certified | Agile (Scrum/Kanban)

4+ Years AI Development, 20+ Years Quality Engineering

PROJECTS & SOLUTIONS

24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering



Java | Cucumber | Kibana | Graylog | Elasticsearch | Container | CI/CD | Mainframe | Dashboard Engineering | Test Automation

Enterprise Legacy Transformation with Fully Automated Quality Assurance The Challenge: Complex mainframe price calculation with millions of combination possibilities needed secure migration - without quality loss or data errors. **The Solution:** 24/7 Dual-Run Validation System with intelligent deviation analysis • **Continuous Testing:** Randomly generated tests across all variations around the clock • **Real-time Monitoring:** Live dashboard with immediate deviation detection on every deployment • **Complete Traceability:** Every deviation uniquely logged with technical and business context • **Automated Root Cause Analysis:** Direct access to calculation paths and error sources • **Quality Intelligence:** Kibana dashboards for end-to-end quality overview **Measurable Impact:** Quality metrics improved from 30% to 96% - secure migration without data loss, complete transparency over migration status

Quality Dashboard - Real-time Overview

Grafana | Elasticsearch | Python | CI/CD | Playwright | Jira API | Confluence API | LeanIX | PDF Generation

Fully automated end-to-end solution for real-time quality monitoring across 10 test tools, 5 environments plus multi-source integration (Pipeline, LeanIX, Jira, Confluence) for automated PDF reports with open issues, bugs, and technical debt. Impact: -80% inquiries, Go/No-Go decisions in minutes instead of hours.

E-Invoicing Automation Platform

Python | Azure Functions | Peppol | Storecove | SAP ByDesign | ZUGFeRD | Factur-X | UBL | Hexagonal Architecture

Fully Automated E-Invoice Processing to EU Standards Core Features: • **Peppol Network Integration**: Connection to European e-invoicing network via Storecove Access Point • **Multi-Format Support**: ZUGFeRD, Factur-X, XRechnung, UBL 2.1, CII (EN 16931 compliant) • **ERP Integration**: Automatic transfer to SAP ByDesign with validation and error handling • **Hexagonal Architecture**: Domain-driven design with Ports & Adapters for maximum testability • **Dual-Channel Inbound**: Storecove Webhook (Push) + Email Inbox (Pull) for redundant processing **Impact:** Production system (v2.1.0) with full EU compliance and 100% automated invoice processing

Email Classification and Processing Workflow

Python | Azure Functions | OpenAI GPT-3.5 | Microsoft Graph API | Elasticsearch | Kibana | Docker

Complete automation solution with Microsoft Graph API, SAP RFC, Azure Functions (Python), OpenAI GPT-3.5-Turbo, Elasticsearch 8.x, Kibana Dashboards, Docker containers, and GitHub Actions CI/CD - 80% reduction in manual work.

AI Context Orchestrator — Multi-Repo Development Platform

Python | YAML | Claude Code Plugins | Hooks API | GitHub Actions | Elasticsearch | Azure Functions | Microsoft Graph API

Central Control System for Enterprise AI Development Core Features: • **Context Orchestrator**: ecosystem.yaml as single source of truth for 20+ repos, customers, infrastructure and skills • **40+ Production Skills**: Log analysis, deployment verification, incident handling, email management, meeting transcription, PDF/certificate generation, SharePoint integration • **Hook-based Permission System**: Smart Guard with project-specific security rules (branch protection, secret detection, deployment gates) • **Structured Work Tracking**: Automatic documentation with daily/weekly archiving and cross-project Kanban board • **Cross-Repo Automation**: Seamless navigation and operations across customer, partner and internal repositories **Impact:** Productivity boost through AI-driven workflows — from incident response to meeting documentation fully automated

AI-Driven Chatbot for Trade-Fair Interaction

Python | OpenAI GPT-4 | Streamlit | FastAPI | Vector Embeddings

Development of an interactive chatbot using RAG (Retrieval-Augmented Generation), OpenAI GPT-4, Vector Embeddings, Python FastAPI, and Streamlit frontend to optimize user interaction on trade-fair websites.

Enterprise AI Consulting: From Strategy to Implementation

AI Strategy | Process Analysis | ROI Assessment | Implementation Planning | Change Management | Executive Workshops | Quality Engineering

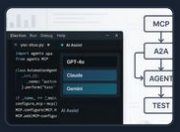
Target Audience: C-Level, Management, Product Owners, Business Analysts **Consulting Services:** • **Strategy Development:** From AI vision to measurable implementation roadmap • **Process Analysis:** Identification of automation potential, quality-first assessment • **ROI Assessment:** Business cases, cost savings, amortization calculations • **Executive Workshops:** Hands-on strategy sessions with live-demo business cases • **Change Management:** Team adoption, training concepts, resistance management • **Implementation Support:** From concept to productive AI solution **Deliverables:** AI strategy template, ROI calculator, implementation roadmap, executive workshop, implementation support

wecation

Flutter | Google Cloud | Firebase | Dart

wecation is a collaborative solution for organising shared experiences: plan, search, coordinate with friends or colleagues and enjoy the activity together! Built with Flutter and Google Cloud.

Developer Workshop: AI-Agent Development & Integration



Anthropic Claude | Google Gemini | OpenAI GPT-4o | MCP | A2A | Playwright | JSON-RPC | IDE Integration | CI/CD

Target Audience: Developers, Technical Leads, DevOps Engineers **Workshop Content:** • **Hands-on AI Agent Development:** Live coding with Claude, Gemini, GPT-4o • **Next-Gen Protocols:** MCP & A2A implementation for enterprise integration • **Browser Automation:** Playwright-MCP for automated test generation • **IDE Integration:** AI assistance in VS Code, Cursor, GitHub Copilot workflows • **Multi-Agent Orchestration:** JSON-RPC, secure agent communication • **Practical Demos:** Jira integration, Elasticsearch queries, GitHub workflows **Deliverables:** Complete source code, slides, hands-on exercises, production-ready templates

Django SLA Dashboard - Enterprise Service Monitoring

Django | Python | PostgreSQL | Chart.js | WebSockets | Docker | REST API

Enterprise SLA Monitoring Dashboard for Service Level Agreements Development of a comprehensive SLA monitoring system as part of a BKS evaluation project (2023-2025) for monitoring Service Level Agreements with real-time metrics and automated alerting. **Core Features:** • **Real-time Monitoring:** 24/7 monitoring of 24+ services with response time tracking (Ø 147ms) • **SLA Compliance:** Automatic calculation and visualization of SLA metrics (99.97% availability) • **Incident Management:** Active incident tracking with medium/low priority classification • **Performance Analytics:** Response time distribution with detailed histograms (<50ms to >500ms) • **Availability Timeline:** 24h history with minute-level resolution and trend analysis • **Alert System:** Proactive alerting for SLA violations and performance degradation **Technical Highlights:** • Django 4.2 with async views for high-performance data processing • PostgreSQL with TimescaleDB extension for time-series data • Django Background Tasks for asynchronous processing • WebSocket integration for live dashboard updates • RESTful API for service integration and CSV export • Docker-based deployment with Kubernetes-ready architecture **Impact:** 60% reduction in MTTR (Mean Time To Repair), proactive incident prevention through trend analysis

QualityCluster

Hugo | JavaScript | GitHub Actions | CI/CD

QualityCluster is an alliance of quality engineers jointly developing better solutions in software quality assurance. Our goal is to implement more quality through a higher degree of automation.

Medical Transcription System

Python | Whisper | Pyannote | Ollama | Llama3 | Apple Silicon | SOAP Protocol

GDPR-Compliant Transcription of Doctor-Patient Conversations Core Features: • **100% Local**: No cloud uploads, all processing on local machine (Privacy-by-Design) • **Speaker Diarization**: Automatic doctor/patient distinction with Pyannote 3.1 (2.6% WER) • **SOAP Protocol Generation**: Automatic creation of structured medical protocols • **Open Source AI**: Ollama with Llama 3.2 for intelligent summaries without API costs • **Apple Silicon Optimized**: Native performance on M1/M2/M3 Macs **Impact:** Healthcare-ready AI solution with full GDPR compliance and zero cloud dependency

AI Learning Platform

Astro | React 19 | TypeScript | Supabase | EU AI Act

EU AI Act Compliant Learning Platform for AI Workshops Core Features: • **Article 4 Compliance**: EU AI Act compliant certifications with audit trail • **15+ Learning Card Types**: Interactive quizzes, exercises, adaptive tests with scoring system • **6 Visual Themes**: Learning, AI-Team, Star Trek, Nerd, Helldivers, Wednesday • **Progress Tracking**: localStorage-based tracking with HR export functionality • **85 Automated Tests**: Quality-assured learning platform **Impact:** Enterprise-ready e-learning for AI governance and structured qualification programs

PROFESSIONAL EXPERIENCE

LLM Infrastructure Architect & Automation Engineer
06/2025 – present

BKS - AI Research & Development

Development of LLM-orchestrated Enterprise Development Ecosystem: Fully integrated system for knowledge management, project automation, and development workflows with equal expertise in Claude, Gemini, and OpenAI GPT.

Tools & Technologies LLM Orchestration: Claude, Google Gemini, OpenAI GPT, GitHub Copilot (equal expertise), Model Context Protocol (MCP), Google Agent Development Kit (ADK) · AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration · Multi-Agent: A2A...

Quality Engineer
09/2025 – 02/2026

TÜV Süd

Quality Engineering for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

Tools & Technologies Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures · Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics · Backend: FastAPI, Python, Pydantic, SQLModel, Azure Functions, Azure Service Bus · Frontend: React, TypeScript, Vite, TanStack...

Platform Quality Architect (CI/CD, Monitoring, Automation)
08/2021 – 05/2025

DVAG

Test management & coordination, as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams. **Key responsibilities** - Design and implementation of functional and technical dashboards with Grafana...

Tools IntelliJ / Visual Studio Code · Java / JavaScript / TypeScript · Playwright / Gauge / Karate · Python for processing and analysis of test and reporting data · CI/CD / GitHub – reusable workflows · Helm / Containers...

Technical Lead & AI Automation Architect
01/2024 – 04/2025

BKS

Technical lead & development: Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform. Full responsibility from requirements analysis through implementation to production go-live.

Tools & technologies Azure Functions: Serverless orchestration of automation workflows. · Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance. · E-Invoicing: Peppol Network, Storecove API,...

AI Engineering Lead & ML Solutions Architect
04/2023 - 04/2024

BKS on behalf of Ryze

Project management & technical lead: Development and deployment of an AI-driven email bot to automate customer communication, AI-assisted processing of incoming emails and triggering of follow-up activities in existing SAP systems. Responsible for end-to-end project coordination and team management from idea to implementation."

Tools & technologies Azure Functions: For serverless application architectures and ML model execution. · Python: Main programming language for the email bot and ML models. · Microsoft Graph API: For accessing and processing emails in Microsoft Outlook 365. · SAP...

DevOps Quality Lead & Dashboard Architect
01/2017 - 05/2021

DB Vertrieb

Test management, test coordination, test automation – agile (Kanban, Scrum, SAFe): Coordination as PO of a cross-team QA team including cross-team onboarding, support, consulting, training **Key responsibilities** - Agile quality assurance through the design of quality gates in the CI/CD pipeline - Design and implementation of functional and technical dashboards...

Tools IntelliJ · Java · Cucumber / Cypress · JMeter / Gatling · CI/CD / GitLab CI / Jenkins · Helm / Docker · JIRA / Confluence · Kibana / Grafana / Instana / Graylog · Behaviour Driven Development ·...

Performance Engineering Lead & Test Architect
09/2015 - 01/2017

DB Systel

Load and performance tests and analysis – test manager, test designer, analyst: Development, execution and analysis of load and performance tests **Key responsibilities** - Requirements management - Status reports - Customer consulting - Documentation in the form of test concepts, analyses, management reporting and wiki documentation - Setup and execution...

Tools Visual Studio · Eclipse · JMeter · Git · JIRA / Confluence · Excel & Access VBS · Interfaces: HTTP, WS, REST, MQ, JMS

FRÜHERE POSITIONEN

AI-driven Automated QA Environment for Energy Infrastructure — AkkuSwap Startup — 07/2024 – 01/2025

Senior iOS Developer & Mobile Architect — Telekom — 04/2015 - 09/2015

Quality Assurance Lead & Test Automation Architect — Siemens — 05/2009 - 03/2015

Performance Test Engineer & Load Testing Specialist — ING-DIBA — 03/2009 - 05/2009

Performance Test Engineer & Quality Consultant — British Telecom, Mobiliar, DB-Systel, Sparkassen Informatik, Loyalty Partner, Telekom, Itelium, Deutsche Post, Postbank — 01/2006 - 12/2008

Workshop: Development with Generative Language Models — Developer Workshop - Agent-based Software Development 06/2025

Presentation: Efficient Documentation through Automation — Enterprise Presentation - AI-powered Documentation Workflows 04/2025

AI Workshop for Business Integration — Applications of AI in Business Context 2023